Program : **Diploma in Biomedical Engineering**	
Course Code : **5249**	Course Title: **Programming in C Lab**
Semester : **5**	Credits: **1.5**
Course Category : **Program Elective**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester : **45**

Course Objectives:

- To provide programming fundamentals using C language.
- To develop logic that will help them to adapt to any other programming language.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic principles and theorems of Engineering Mathematics		Mathematics I &II	1 & 2
Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Make use of various data types, I/O statements and basic operators in C programming	11	Applying
CO2	Apply decision making and loop statements in C programming.	13	Applying
CO3	Construct C programs based on arrays.	9	Applying
CO4	Develop C programs using string handling functions and user-defined functions	9	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Make use of various data types, I/O statements and basic operators in C programming**		
M1.01	Demonstrate output and input functions.	5	Understanding
M1.02	Demonstrate different fundamental data types in C.	3	Understanding
M1.03	Solvesimple problem statements using operators.	3	Applying
CO2	**Apply decision making and loop statements in C programming**		
M2.01	Solve problem statements in C using flow control statements - if, if … else, nested if, switch.	8	Applying
M2.02	Solve problem statements using loop statements - while, do … while and for loop.	5	Applying
	Series Test 1	1.5	
CO3	**Construct C programs based on arrays.**		
M3.01	Make use of one dimensional array operations - largest, smallest and compute sum & average of array elements	3	Applying
M3.02	Write programs to perform two dimensional array operations - transpose of matrix, matrix addition, matrix subtraction	6	Applying
CO4	**Develop C programs using string handling functions and user-defined functions.0**		
M4.01	Implement string manipulation functions - strcpy( ), strcat( ), strlen( ), strcmp( ).	3	Applying
M4.02	Illustrate ‘User-defined functions’	6	Applying
	Series Test 2	1.5	

**** - Suggested Open Ended Projects**

(Students have to do open ended experimentsduring the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.)

Example:

- Develop program such as
- Student CGPAcalculator,
- Consumer billing,
- Candidate list sorting

Text / Reference:

T/R	Book Title/Author
T1	Let Us C – YashavantKanetkar, BPB Publications
R2	Programming in ANSI C, E. Balagurusamy, Tata McGraw Hill
R3	A Text Book on C, E. Karthikeyan , PHI Learning Pvt. Ltd
R4	Programming in C D. Ravichandran, New Age International Publishers
R5	C. Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall

Online Resources:

Sl.No	Website Link
1	http://freecomputerbooks.com/The-C-Programming-Language.html
2	https://www.tutorialspoint.com/cprogramming/index.htm
3	https://www.guru99.com/c-programming-tutorial.html
4	https://www.cprogramming.com/
5	http://www.zentut.com/c-tutorial/

Sample Questions to Test Outcomes

1. Implement a C program to find sum and average of all element sin an integer array of size MxN where M,N and array elements can be input while the program isexecuting.
2. Implement a program to check whether the given string is a Palindrome or not.
3. Write a program to accept a string and count the number of vowels present in this string